

MALWARE INTELLIGENCE

Yellow RAT Malware Intelligence Investigation

Hash enrichment, threat-cluster identification, host artifacts, and command-and-control analysis

EXECUTIVE CONCLUSION

The supplied SHA-256 was associated with a malicious DLL tracked by Red Canary as Yellow Cockatoo. The investigation identified a persistent host artifact and historical C2 infrastructure without executing the sample.

Report field	Value
Analyst	Alaa Zahra
Platform	CyberDefenders
Investigation type	Static malware intelligence and OSINT enrichment
Evidence	SHA-256 supplied in hash.txt
Report date	07 October 2026
Time standard	UTC
Sample execution	Not performed

Handling notice The C2 domain is defanged throughout this report. Do not browse to historical malicious infrastructure.

Executive Summary

This investigation began with a single SHA-256 value recovered from a small evidence archive. VirusTotal enrichment linked the indicator to a DLL sample with a strong malicious detection consensus. Because vendor labels were inconsistent, the investigation pivoted to Red Canary technical reporting, which identifies the associated threat cluster as Yellow Cockatoo.

The evidence also revealed a commonly observed filename, PE compilation metadata, the first recorded VirusTotal submission, an AppData artifact used as a host identifier, and historical command-and-control infrastructure. These findings provide defenders with concrete hunting pivots while preserving the distinction between directly observed metadata and broader cluster behavior.

Key Findings

Finding	Result	Source
Threat cluster	Yellow Cockatoo	Red Canary
SHA-256	30e527e4...10d071c85	Supplied evidence / VirusTotal
Filename	111bc461-1ca8-43c6-97ed-911e0e69fdf8.dll	VirusTotal
Compilation time	2020-09-24 18:26:47 UTC	VirusTotal PE metadata
First VT submission	2020-10-15 02:47:37 UTC	VirusTotal history
Host artifact	%APPDATA%\solarmarker.dat	Red Canary
C2 domain	gogohid[.]com	Red Canary

Analytic Confidence

- High confidence in the SHA-256 because it was supplied directly and matched the VirusTotal record.
- High confidence in the threat-cluster name, host artifact, and C2 domain because they are documented by Red Canary technical analysis for the matching activity.
- Medium confidence in the filename as a durable hunting indicator because filenames can be changed without altering file content.

Priority Actions

- Hunt the SHA-256 across endpoint, email, download, and file inventory telemetry.
- Search endpoints for %APPDATA%\solarmarker.dat and reconstruct its creation timeline.
- Review DNS, proxy, firewall, and EDR events for the defanged domain and documented URI paths.
- Isolate confirmed affected endpoints and preserve memory, suspicious files, process trees, and persistence artifacts.

Scope and Methodology

Objective

The objective was to turn a supplied hash into defensible malware intelligence: identify the threat cluster, recover file and timeline metadata, identify host and network artifacts, and translate the results into detection and response guidance.

Evidence Handling

The investigation was performed in an isolated Kali Linux virtual machine. The archive was listed before extraction. The evidence contained a text file holding a SHA-256 value rather than an executable sample, so no malware was launched or dynamically detonated.

Sources and Tools

Source or tool	Purpose
7-Zip / terminal	Inspect archive structure and read the supplied hash
VirusTotal	Enrich the hash with detection, filename, PE, and history metadata
Red Canary analysis	Resolve the named threat cluster and recover behavioral artifacts
Hybrid Analysis	Attempt historical lookup; anonymous time-range restriction limited results

Evidence Standard

- Direct observation: values visible in the supplied file or VirusTotal record.
- Corroborated intelligence: behavior documented by a reputable technical report linked to the threat cluster.
- Inference: conclusions derived from timing or behavior and explicitly qualified as such.

EVIDENCE CAUTION

A compilation timestamp is not an execution timestamp, and a VirusTotal first-submission timestamp is not the first infection date. Both are useful timeline clues only.

Evidence Triage

The archive was inspected before opening its contents. This established that the lab provided an indicator for enrichment rather than a binary for local analysis.

```
7z l 127-Yellow-RAT.zip -pcyberdefenders.org
```

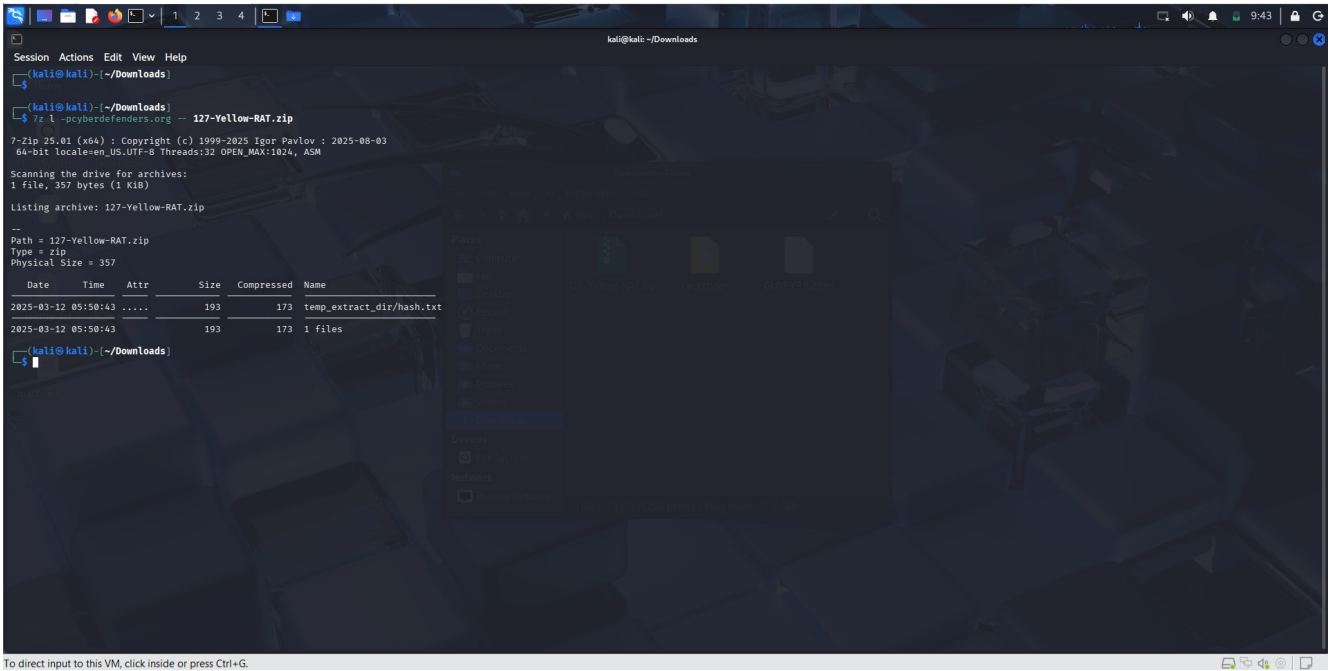


Figure 1 — Archive listing showing the supplied hash.txt artifact.

The text file contained one 64-character hexadecimal value, consistent with a SHA-256 digest. This became the stable pivot used across intelligence sources.

```
30e527e45f50d2ba82865c5679a6fa998ee0a1755361ab01673950810d071c85
```

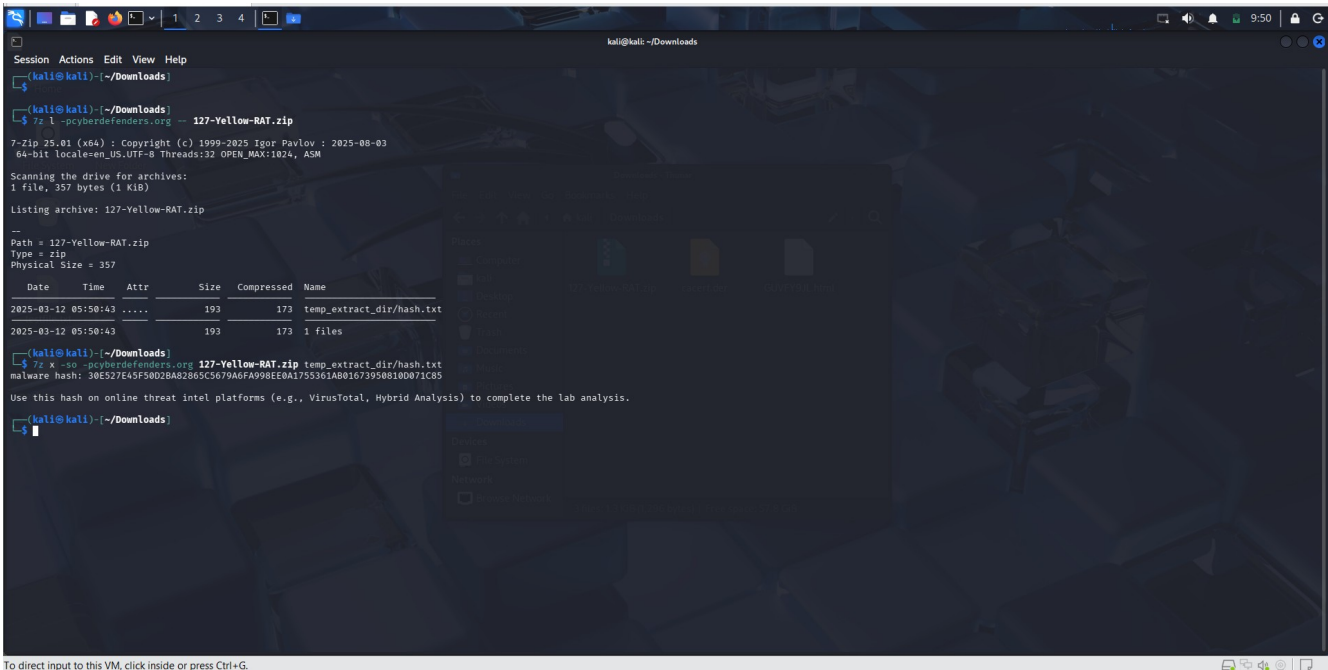


Figure 2 — The SHA-256 recovered from the supplied evidence.

VirusTotal Enrichment

VirusTotal associated the SHA-256 with a DLL and showed 56 of 71 engines detecting it as malicious at the time of analysis. This provides a strong prioritization signal, but the ratio alone does not prove attribution or establish one canonical family name.

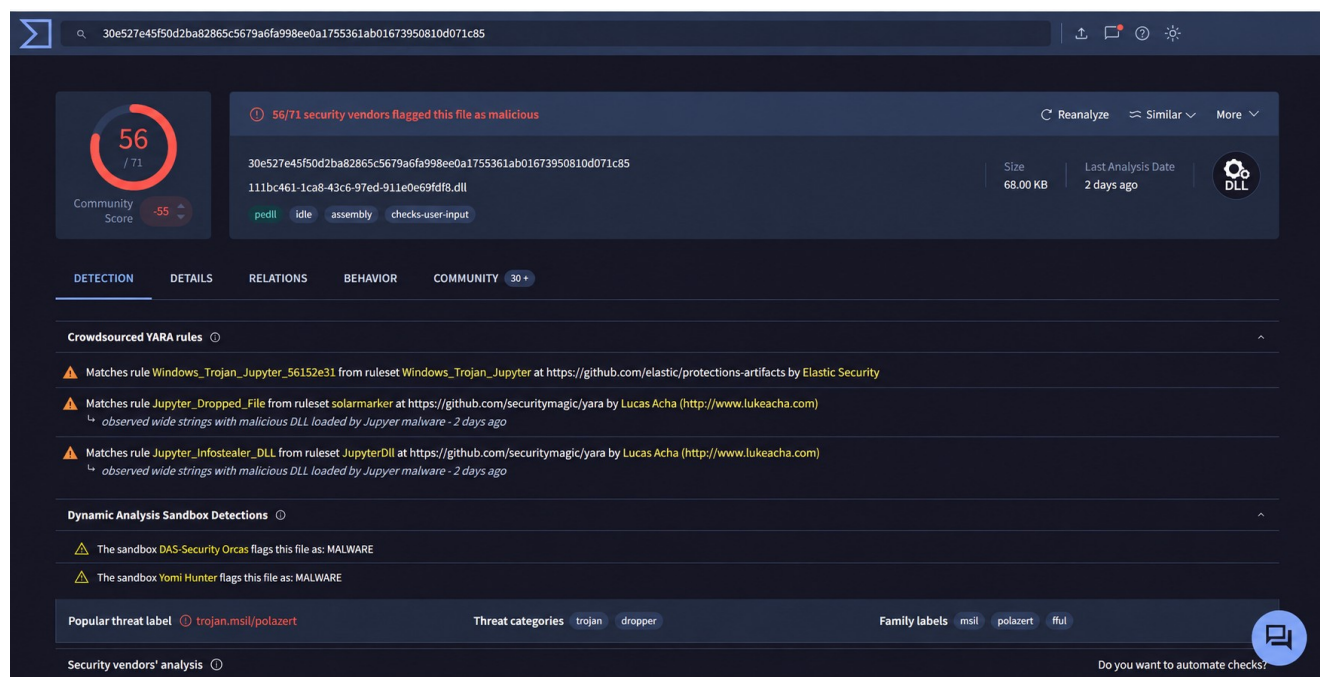


Figure 3 — VirusTotal detection overview for the supplied SHA-256.

Vendor Naming Differences

The engines used several labels for the same sample. Such differences can reflect vendor-specific taxonomies, overlapping clusters, generic detections, or different intelligence update schedules. The question named Red Canary, so the analysis pivoted to Red Canary's own technical publication instead of selecting the most frequent vendor label.

30e527e45f50d2ba82865c5679a6fa998ee0a1755361ab01673950810d071c85

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	Trojan.Win32.Tiggre.C4226862	Alibaba	Trojan:MSIL/Polazert.ed87dff8
AliCloud	Trojan:MSIL/Injector.J	ALYac	Trojan:MSIL.Polazer
Antiy-AVL	Trojan/Win32.Agent	Arcabit	Trojan.Agent.FFUL
Arctic Wolf	Unsafe	Avast	Win32:MalwareX-gen [Misc]
AVG	Win32:MalwareX-gen [Misc]	Avira (no cloud)	TR/Dropper.Gen7
BitDefender	Trojan.Agent.FFUL	Bkav Pro	W32.Malware.8070A725
ClamAV	Win.Trojan.Jupiter-9858780-0	CrowdStrike Falcon	Win/malicious_confidence_100% (W)
CTX	DLL.trojan.msil	DeepInstinct	MALICIOUS
DrWeb	Trojan.Polazert.1	Elastic	Windows.Trojan.Jupyter
Emsisoft	Trojan.Agent.FFUL (B)	eScan	Trojan.Agent.FFUL
ESET-NOD32	MSIL/Polazert.B Trojan	Fortinet	W32/Generic.Bitr
GData	Trojan.Agent.FFUL	Google	Detected
Gridinsoft (no cloud)	Trojan.Win32.Heur.oals1	Huorong	Trojan/MSIL.Agent.y
Ikarus	Backdoor.VenomRAT	Jiangmin	Trojan.Generic.ghect
K7AntiVirus	Trojan (700000201)	K7GW	Trojan (700000201)

Figure 4 — Divergent vendor labels for the same sample.

Threat Cluster Identification

Red Canary identifies the activity cluster as Yellow Cockatoo. This term is an analytic cluster label. It should not be presented as a verified real-world identity, and it may overlap with activity that other vendors call Jupyter or SolarMarker.

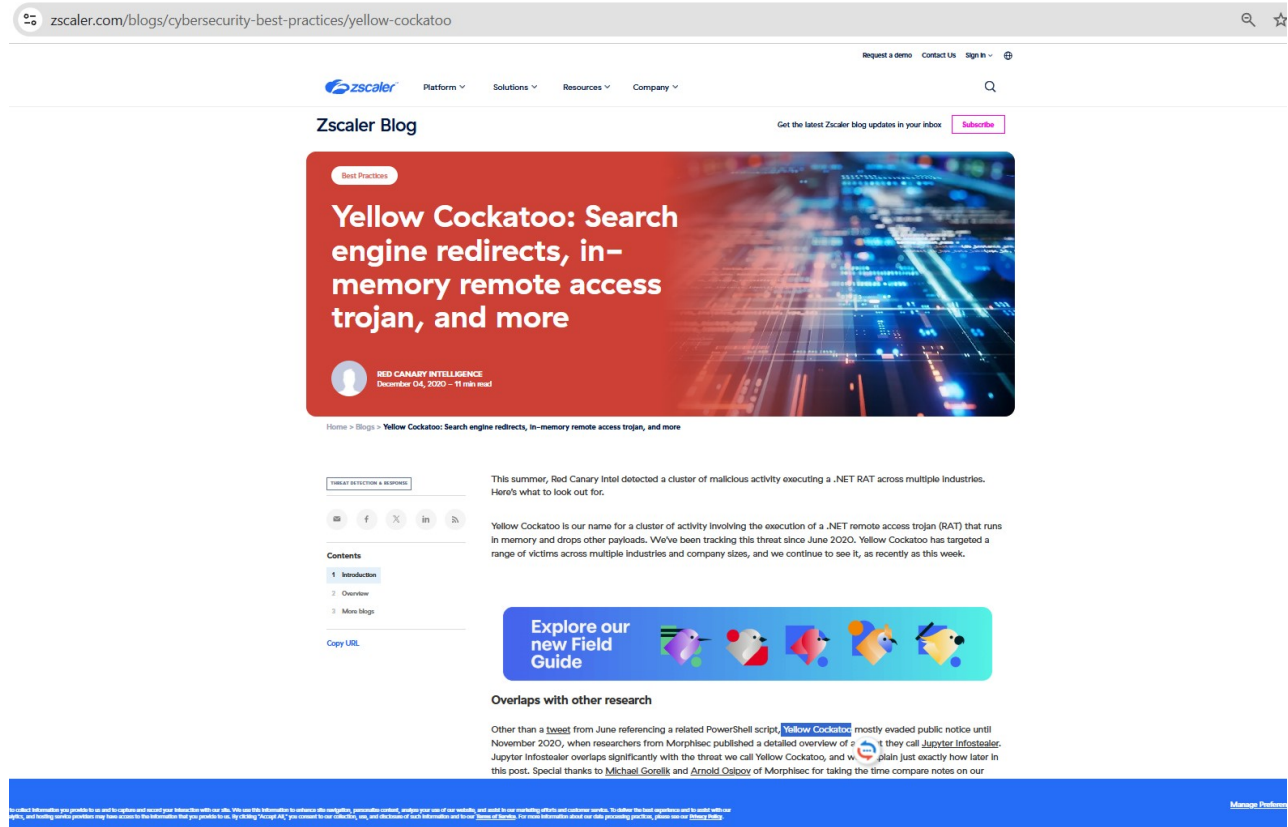


Figure 5 — Red Canary technical analysis identifying Yellow Cockatoo.

ATTRIBUTION NOTE

Threat-intelligence names describe how an organization groups related activity. They are not automatically interchangeable, and they do not by themselves identify the human operator.

Why the Source Matters

When a question asks for the name used by a particular organization, the authoritative evidence is that organization's technical analysis. VirusTotal remains valuable for cross-vendor context, but it aggregates labels rather than deciding which naming convention is correct.

File and Timeline Metadata

Associated Filename

VirusTotal listed the filename 111bc461-1ca8-43c6-97ed-911e0e69fdf8.dll. It is useful for threat hunting, but it should be correlated with the hash and behavior because attackers can rename files.

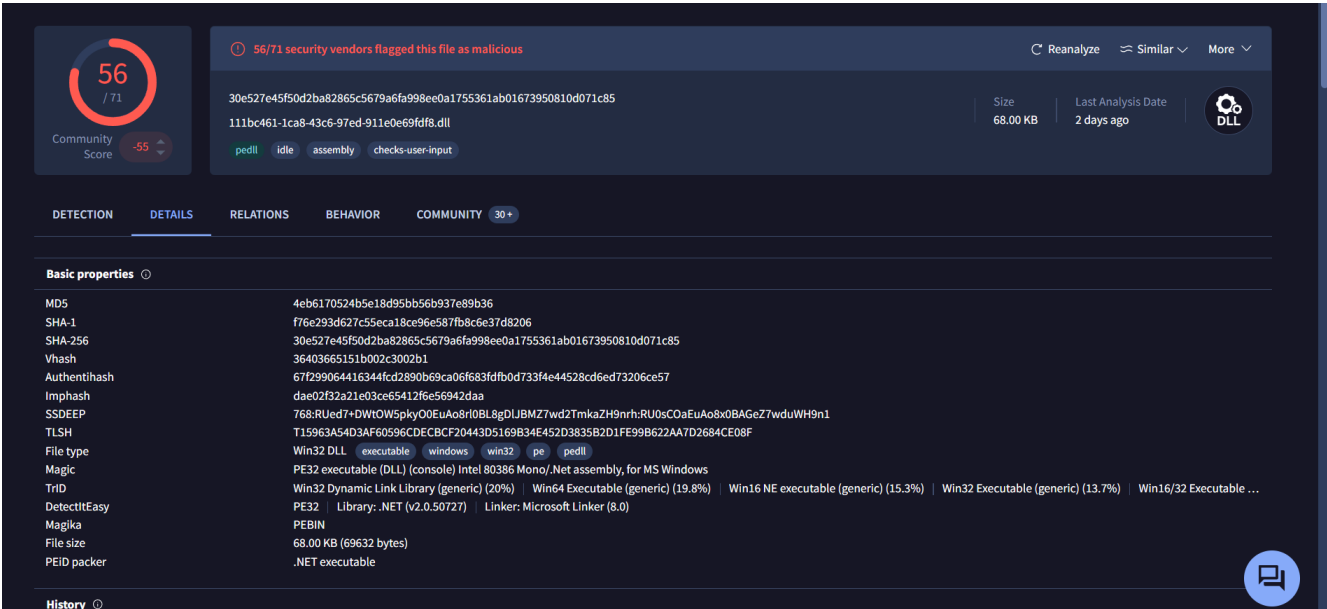


Figure 6 — VirusTotal properties showing the associated filename and hash.

Compilation Timestamp

The PE header recorded 2020-09-24 18:26:47 UTC as the compilation timestamp. This may indicate when the binary was built, but PE timestamps are attacker-controlled metadata and can be altered.

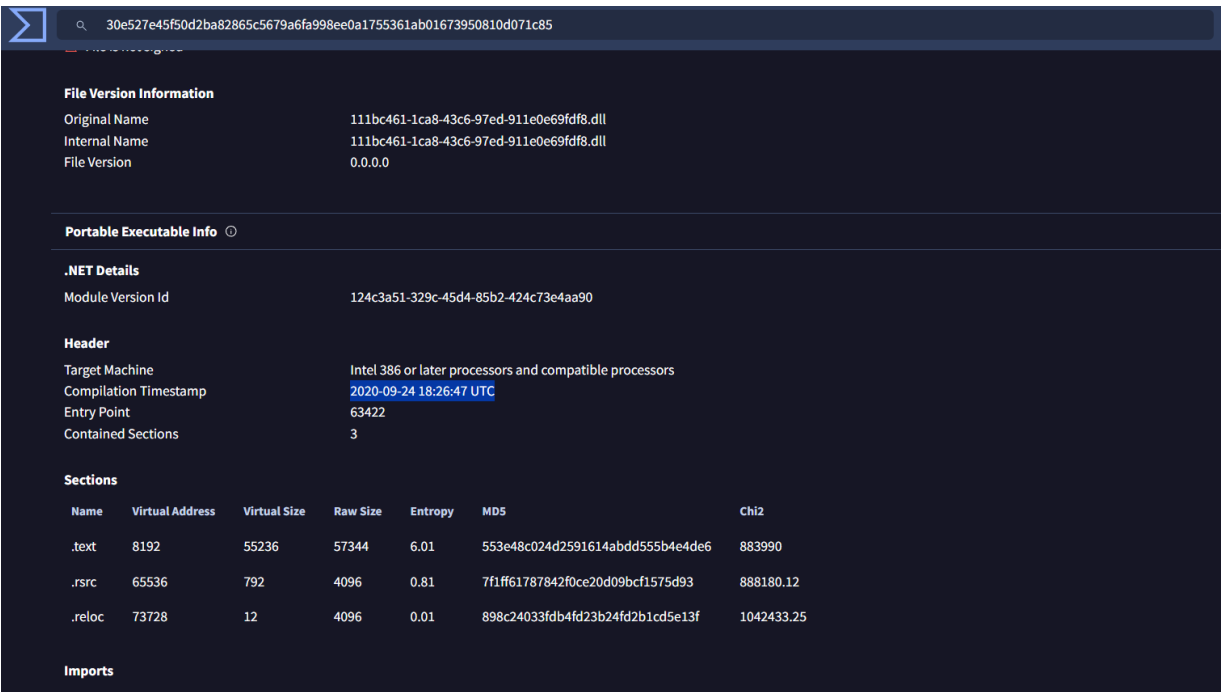


Figure 7 — PE compilation timestamp displayed by VirusTotal.

VirusTotal Submission History

VirusTotal recorded the first submission at 2020-10-15 02:47:37 UTC. The approximately 20-day gap from the compilation timestamp can guide timeline questions, but it does not establish when the sample first infected a system or first appeared in the wild.

The screenshot shows the VirusTotal interface for a file with hash 30e527e45f50d2ba82865c5679a6fa998ee0a1755361ab01673950810d071c85. The file is identified as a Win32 DLL, specifically a PE32 executable (DLL) (console) Intel 80386 Mono/.Net assembly, for MS Windows. The file size is 68.00 KB. The submission history shows the first submission on 2020-10-15 02:47:37 UTC, which is highlighted in blue. The last submission was on 2026-05-01 16:46:44 UTC, and the last analysis was on 2026-10-05 12:01:42 UTC. The names section lists several related file names, including 111bc461-1ca8-43c6-97ed-911e0e69fdf8.dll, 30e527e45f50d2ba82865c5679a6fa998ee0a1755361ab01673950810d071c85, and 4eb6170524b5e18d95bb56b937e89b36.dll.

Figure 8 — VirusTotal history showing the first-submission timestamp.

Timestamp	Meaning	What it does not prove
2020-09-24 18:26:47 UTC	Timestamp stored in the PE header	First execution or first compromise
2020-10-15 02:47:37 UTC	First recorded upload to VirusTotal	First appearance in the wild

Host Artifact Analysis

The Red Canary analysis documents a file created in the roaming AppData directory: %USERPROFILE%\AppData\Roaming\solarmarker.dat. The malware uses this file to store a randomly generated identifier for the infected host.

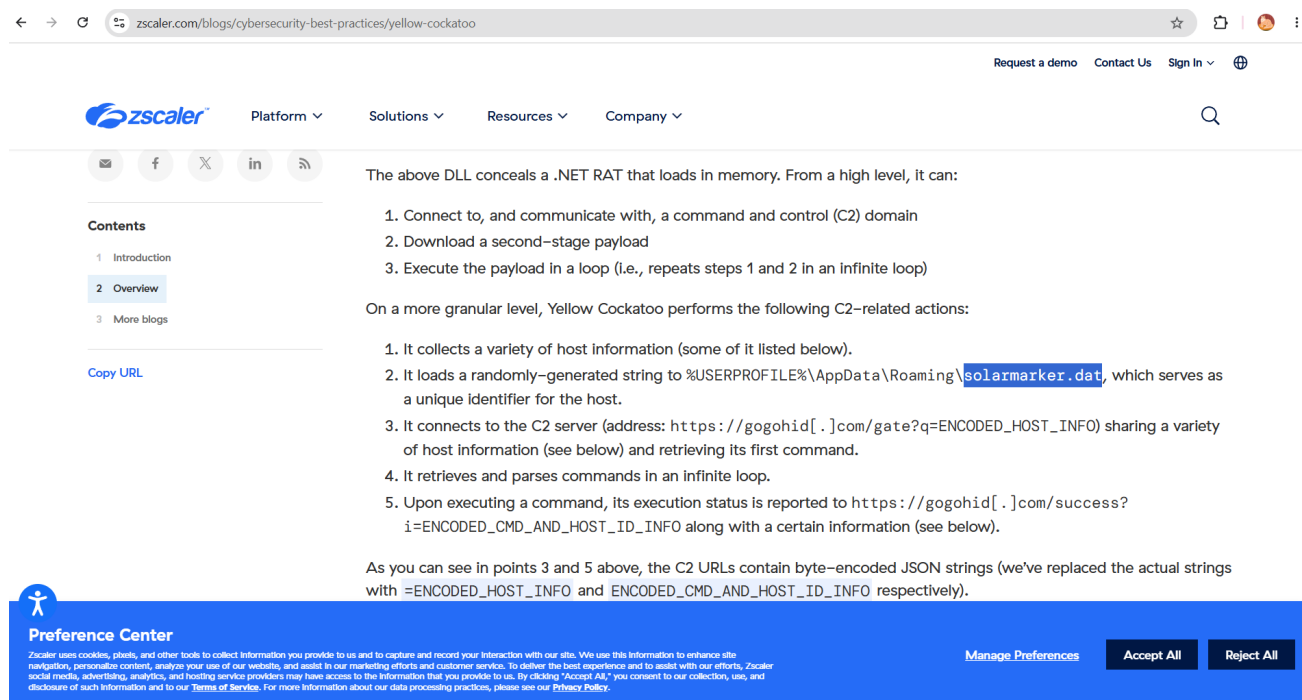


Figure 9 — Technical analysis describing solarmarker.dat and its purpose.

Defensive Value

- The path provides a concrete endpoint hunting pivot beyond the original DLL hash.
- File creation telemetry can identify the responsible process and nearby activity.
- The artifact should be correlated with persistence, PowerShell, process injection, and network events before declaring an incident.

Suggested Hunt

Search file inventory and EDR events for filenames equal to solarmarker.dat under user Roaming AppData directories. For each match, capture the file hash, creation time, parent process, user context, and related outbound connections.

Command and Control Analysis

The technical report links the sample to the historical C2 domain gogohid[.]com. It describes /gate as a path used for sending host information and retrieving commands and /success as a path used to report command execution status.

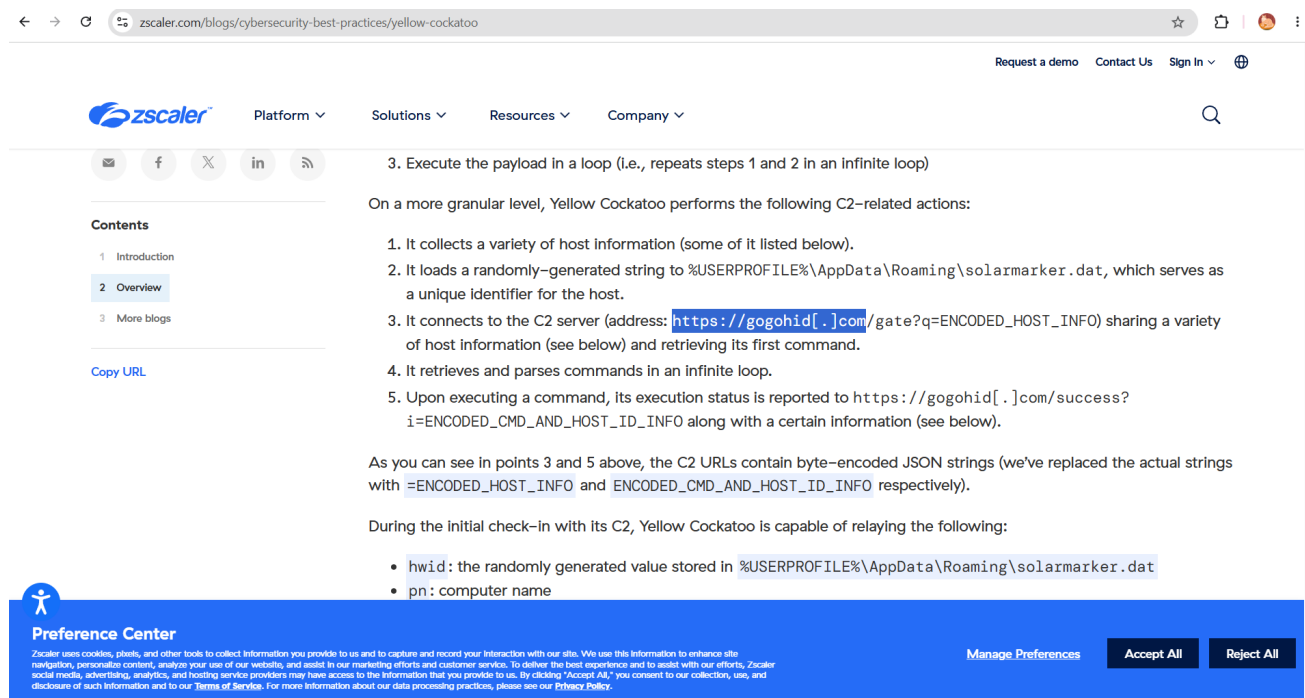


Figure 10 — Technical analysis describing the historical C2 workflow.

Network Hunting Guidance

- Search DNS records for resolutions of the defanged domain.
- Search proxy and firewall logs for connections to resolved IP addresses around the suspected period.
- Review HTTP or TLS metadata for the documented URI paths where visibility permits.
- Correlate network hits with the SHA-256, DLL filename, solarmarker.dat, and suspicious process trees.

OPERATIONAL CAUTION

Historical infrastructure can expire, change ownership, or be repurposed. Validate current reputation and business context before applying permanent controls.

Indicators of Compromise

Type	Indicator	Confidence	Analyst note
SHA-256	30e527e45f50d2ba82865c5679a6fa998ee0a1755361ab01673950810d071c85	High	Stable content identifier
Filename	111bc461-1ca8-43c6-97ed-911e0e69fdf8.dll	Medium	Renamable; correlate with hash
File path	%APPDATA%\solarmarker.dat	High	Host identifier artifact
Domain	gogohid[.]com	High	Historical C2; validate current status
URI	/gate	Medium	Host information and commands
URI	/success	Medium	Execution result reporting

MITRE ATT&CK Mapping

These mappings describe behavior documented for the broader Yellow Cockatoo cluster. Additional endpoint or network telemetry is required before asserting that every technique occurred on a specific host.

Technique	ID	Evidence rationale
Ingress Tool Transfer	T1105	Downloads additional payloads
Application Layer Protocol Web Protocols	T1071.001	HTTPS-based C2 communication
System Information Discovery	T1082	Collects and sends host information
PowerShell	T1059.001	PowerShell-based execution described in cluster reporting
Process Hollowing	T1055.012	Documented rpe command performs process hollowing
Registry Run Keys Startup Folder	T1547.001	Startup shortcut persistence described in reporting

Detection and Response Plan

Immediate Containment

- Isolate endpoints with corroborated detections while preserving evidence.
- Block confirmed malicious infrastructure at DNS, proxy, and firewall layers where appropriate.
- Quarantine matching files and prevent execution using EDR controls.

Enterprise Hunt

- Query endpoint telemetry for the SHA-256 and associated filename.
- Search for %APPDATA%\solarmarker.dat across user profiles.
- Review Startup folders and shortcut creation for persistence.
- Correlate PowerShell, process injection, and unusual child-process behavior.
- Review DNS and proxy logs for the historical C2 and related destinations.

Evidence Collection

- Capture volatile memory before shutdown when feasible.
- Preserve suspicious files, file-system metadata, process trees, command lines, and network telemetry.
- Record timezone and clock accuracy before reconstructing the incident timeline.
- Retain original evidence separately from working copies and document hashes for acquired files.

Recovery and Monitoring

- Remove verified persistence and rebuild systems when trust cannot be restored.
- Reset credentials only when investigation evidence or exposure scope supports the action.
- Deploy temporary detections for the observed indicators and monitor for recurrence.
- Review defensive gaps that allowed the initial file to reach or execute on endpoints.

Limitations and Lessons Learned

Limitations

- The lab supplied a hash rather than the executable, so no local static disassembly or dynamic execution was performed.
- VirusTotal labels represent multiple vendor taxonomies and are not authoritative attribution.
- VirusTotal behavior data did not expose every requested artifact for this sample.
- Hybrid Analysis restricted older searches for anonymous access; the investigation pivoted to the authoritative Red Canary report.
- Compilation and first-submission timestamps do not establish victim infection time.

Transferable Method

- Identify what the evidence actually is before choosing the analysis path.
- Use cryptographic hashes as stable pivots and filenames as supporting clues.
- Resolve vendor naming conflicts through the source organization's own reporting.
- Separate direct observation, corroborated intelligence, and inference in every conclusion.
- When a source lacks an artifact, pivot to another reputable source and record the limitation.

Conclusion

The investigation converted a single SHA-256 into a defensible set of malware-intelligence findings. The sample is associated with activity tracked by Red Canary as Yellow Cockatoo, with useful file, host, timeline, and network indicators for enterprise hunting. The strongest operational approach is correlation: no single filename, timestamp, or domain should be treated in isolation.

References

- VirusTotal sample report: <https://www.virustotal.com/gui/file/30e527e45f50d2ba82865c5679a6fa998ee0a1755361ab01673950810d071c85>
- Red Canary: <https://redcanary.com/blog/threat-intelligence/yellow-cockatoo/>
- Zscaler-hosted technical analysis: <https://www.zscaler.com/blogs/cybersecurity-best-practices/yellow-cockatoo>
- MITRE ATT&CK Enterprise: <https://attack.mitre.org/>

DISCLAIMER

This report documents an authorized training investigation. Indicators should be validated against current intelligence and local telemetry before operational use.